Bayesian Inference for the Cox–Ingersoll–Ross Process

The stationary law, a stationary percentile, and a conjugate posterior for mean reversion

Angelo Saporito

October 2026

The Cox–Ingersoll–Ross (CIR) process is the variance process in Heston's stochastic volatility model and a standard model of short-term interest rates. These notes work out three pieces of a Bayesian treatment. The stationary law is Gamma, and it identifies the long-run mean and the ratio of the shock variance to the speed of mean reversion, but not the speed itself. A percentile of the stationary law can be estimated from the marginal distribution of the data, but standard errors that treat a sampled path as independent draws are badly overconfident. The speed of mean reversion is identified only by transitions: after a square-root transformation and an Euler discretization on a fine grid, the drift parameters have a Gaussian conditional posterior given the path, which is the core block of a data-augmentation sampler. Simulations check each result.

Written up in October 2026 from handwritten notes made in 2024–25, with corrections. The revision notes list what changed.

1 Setup

The CIR process solves

$$dv_t = \kappa(\theta - v_t)\,dt + \sigma\sqrt{v_t}\,dW_t, \qquad \kappa, \theta, \sigma > 0, \tag{1}$$

where θ is the long-run mean, κ the speed of mean reversion, and σ scales the shocks. In Heston's model v_t is the instantaneous variance of an asset's returns, and $\sqrt{v_t}$ is its volatility (Heston 1993); the same process is a model of the short-term interest rate (Cox et al. 1985).

Throughout, v_t is treated as observed at discrete dates. That fits interest rates directly. For Heston's model it is a simplification: there the variance is latent, and a full treatment would add the price equation (Section 6). The drift parameters are collected as

$$\psi = (\psi_1, \psi_2) = (\kappa\theta,\ \kappa),$$

so that the drift $\kappa(\theta - v) = \psi_1 - \psi_2 v$ is linear in ψ. The running example uses $\kappa = 2$, $\theta = 0.04$, and $\sigma = 0.25$, a long-run volatility of 20%.

2 The stationary law and what it identifies

The companion note on the Fokker–Planck equation (Saporito 2026, sec. 3.3) derives the stationary law of Equation 1 from the zero-flux condition $\kappa(\theta - v)\varphi = \frac{1}{2}\frac{d}{dv}[\sigma^2 v\varphi]$:

$$v_\infty \sim \text{Gamma}(\alpha, \beta), \qquad \alpha = \frac{2\kappa\theta}{\sigma^2}, \qquad \beta = \frac{2\kappa}{\sigma^2}, \tag{2}$$

with shape α, rate β, mean θ, and variance $\theta\sigma^2/2\kappa$. The process stays strictly positive when the Feller condition $2\kappa\theta \geq \sigma^2$ holds, which is the same as $\alpha \geq 1$ (Feller 1951). In the running example $\alpha = 2.56$ and $\beta = 64$, and the condition holds.

The stationary law depends on the three parameters only through two combinations. Its mean, $\alpha/\beta = \theta$, identifies the long-run level, and its rate, $\beta = 2\kappa/\sigma^2$, identifies the ratio σ^2/κ. Scaling κ and σ^2 by a common factor leaves Equation 2 unchanged: a process that reverts twice as fast, hit by shocks with twice the variance, has exactly the same long-run distribution. The marginal distribution of the data can therefore deliver θ and any function of the stationary law, such as its percentiles, but it cannot deliver the speed of mean reversion. That requires the transitions, which are the subject of Section 4.

3 A stationary percentile

Risk calculations often need a high percentile of the stationary distribution: the level of variance exceeded only 5% of the time in the long run. Write $q = F^{-1}(0.95; \alpha, \beta)$ for the 95th percentile of the Gamma law. Because the square root is increasing, the 95th percentile of stationary volatility is $\sqrt{q}$. In the running example $q = 0.0879$, a volatility of 29.7%.

3.1 Maximum likelihood and the delta method

Suppose first that $x_1, \ldots, x_n$ are independent draws from the stationary law. Section 3.2 returns to the case of a single sampled path. The log-likelihood for Gamma data with shape a and rate b is

$$\ell(a, b) = na \log b - n \log \Gamma(a) + (a - 1) \sum_{i=1}^{n} \log x_i - b \sum_{i=1}^{n} x_i.$$

Setting the derivative in b to zero gives $\hat{b} = \hat{a}/\bar{x}$, and substituting into the derivative in a gives the equation

$$\log \hat{a} - \text{digamma}(\hat{a}) = \log \bar{x} - \frac{1}{n} \sum_{i=1}^{n} \log x_i,$$

where digamma is the derivative of $\log\Gamma$; the equation has a unique solution, found by Newton's method. The second derivatives are $\ell_{aa} = -n\,\text{trigamma}(a)$, $\ell_{bb} = -na/b^2$, and $\ell_{ab} = n/b$, where trigamma is the derivative of digamma. The Fisher information is minus their expectation, and since its determinant is $n^2(a\,\text{trigamma}(a) - 1)/b^2$, its inverse is

$$\mathcal{J}^{-1}(a, b) = \frac{1}{n(a\,\text{trigamma}(a) - 1)} \begin{pmatrix} a & b \\ b & b^2\,\text{trigamma}(a) \end{pmatrix}. \tag{3}$$

The denominator is positive because trigamma$(a) > 1/a$ for every $a > 0$.

The percentile depends on (a, b) through $q = Q(a)/b$, where $Q(a)$ is the 95th percentile of the Gamma law with rate one. Hence $\partial q/\partial b = -q/b$, and $\partial q/\partial a = Q'(a)/b$ can be computed by a central finite difference. The delta method then gives

$$\mathrm{se}(\hat{q})^2 = g^\top \mathcal{J}^{-1}(\hat{a}, \hat{b})\, g, \qquad g = \left(\frac{\partial q}{\partial a},\ \frac{\partial q}{\partial b}\right)^\top,$$

and the interval $\hat{q} \pm 1.96\ \mathrm{se}(\hat{q})$.

A simulation with $n = 500$ draws from the running example's stationary law, repeated 20,000 times, checks the formula. The standard deviation of $\hat{q}$ across samples is 0.00292, against a delta-method standard error of 0.00291, and the 95% interval covers the true percentile in 94.8% of samples.

3.2 Two cautions

First, the percentile should be computed from the Gamma law itself. The normal approximation $(a + 1.645\sqrt{a})/b$, mean plus 1.645 standard deviations, gives 0.0811 in the running example, 7.7% below the true value, because the Gamma law is skewed to the right.

Second, a sampled path is not a set of independent draws. Consecutive observations of a CIR process are strongly correlated (the autocorrelation at lag δ is $e^{-\kappa\delta}$), so a long daily series carries far less information about the stationary law than its length suggests.

Applied to ten years of daily data from the running example (2,520 observations, 2,000 simulated paths), the formula above reports a standard error of 0.0013, while the actual standard deviation of $\hat{q}$ across paths is 0.0170: the independence assumption understates the uncertainty by a factor of 13.1. A block bootstrap would repair the interval. The Bayesian route of Section 4.5 handles the dependence through the transition density instead.

4 Learning the speed of mean reversion from transitions

4.1 The square-root transformation

The diffusion coefficient in Equation 1 depends on the state. Transforming to $y_t = \sqrt{v_t}$ makes it constant. With $g(v) = \sqrt{v}$, $g'(v) = 1/(2\sqrt{v})$, and $g''(v) = -1/(4v^{3/2})$, Itô's formula gives

$$dy_t = g'(v_t)\, dv_t + \tfrac{1}{2} g''(v_t)\, (dv_t)^2 = \frac{\kappa(\theta - y_t^2)}{2y_t}\, dt + \frac{\sigma}{2}\, dW_t - \frac{\sigma^2}{8y_t}\, dt,$$

so that

$$dy_t = \left[\frac{\psi_1 - \sigma^2/4}{2y_t} - \frac{\psi_2\, y_t}{2}\right] dt + \frac{\sigma}{2}\, dW_t. \tag{4}$$

The term $-\sigma^2/4$ is the Itô correction. It matters below: it shifts the posterior mean of $\kappa\theta$ by exactly $\sigma^2/4$.

4.2 Discretization and data augmentation

Suppose v is observed at T dates spaced δ apart, for example daily. The Euler discretization of Equation 4 is accurate only over short steps, so between each pair of consecutive observations we insert M unobserved points, giving a fine grid with spacing $\Delta = \delta/(M+1)$ and points $y_0, y_1, \dots, y_N$, where $N = (T-1)(M+1)$ is the number of transitions (Eraker 2001). On this grid the Euler step is

$$y_{j+1} = y_j + \left[\frac{\psi_1 - \sigma^2/4}{2y_j} - \frac{\psi_2\, y_j}{2}\right]\Delta + \frac{\sigma}{2}\sqrt{\Delta}\,\varepsilon_j, \qquad \varepsilon_j \sim N(0,1) \text{ independent.}$$

Moving the Itô term to the left-hand side turns each step into a linear regression with known error variance:

$$z_j = x_j^\top \psi + e_j, \qquad z_j = (y_{j+1} - y_j) + \frac{\sigma^2\Delta}{8y_j}, \qquad x_j = \begin{pmatrix} \Delta/(2y_j) \\ -y_j\Delta/2 \end{pmatrix}, \qquad e_j \sim N\!\left(0, \frac{\sigma^2\Delta}{4}\right). \tag{5}$$

4.3 The conditional posterior

Given the complete path and σ, the likelihood of ψ is a product of Gaussian densities with variance $\sigma^2\Delta/4$:

$$p(y_{1:N} \mid y_0, \psi, \sigma) \;\propto\; \exp\left\{-\frac{2}{\sigma^2\Delta}\sum_{j=0}^{N-1}\left(z_j - x_j^\top\psi\right)^2\right\}.$$

Expanding the square, with $S = \sum_j x_j x_j^\top$ and $s = \sum_j x_j z_j$,

$$\sum_j \left(z_j - x_j^\top\psi\right)^2 = \psi^\top S\psi - 2\,\psi^\top s + \sum_j z_j^2.$$

Under a flat prior the posterior is therefore

$$\pi(\psi \mid y_{0:N}, \sigma) \;\propto\; \exp\left\{-\tfrac{1}{2}\,\psi^\top\Lambda\,\psi + \psi^\top h\right\}, \qquad \Lambda = \frac{4}{\sigma^2\Delta}\,S, \qquad h = \frac{4}{\sigma^2\Delta}\,s.$$

The factor is $4/(\sigma^2\Delta)$, not $2/(\sigma^2\Delta)$, because the Gaussian kernel is written with a factor of one half in front of the quadratic form. Completing the square,

$$-\tfrac{1}{2}\,\psi^\top\Lambda\,\psi + \psi^\top h = -\tfrac{1}{2}(\psi - \Lambda^{-1}h)^\top\Lambda(\psi - \Lambda^{-1}h) + \text{const},$$

so

$$\psi \mid y_{0:N}, \sigma \;\sim\; N(\Lambda^{-1}h,\; \Lambda^{-1}). \tag{6}$$

This is the least-squares fit of Equation 5, $\hat{\psi} = (X^\top X)^{-1}X^\top z$, with covariance $(\sigma^2\Delta/4)(X^\top X)^{-1}$.

The sums reduce to four statistics of the path,

$$A = \sum_j \frac{1}{y_j^2}, \qquad B = \sum_j y_j^2, \qquad C = \sum_j \frac{y_{j+1} - y_j}{y_j}, \qquad D = \sum_j y_{j+1}\,y_j,$$

all taken over $j = 0, \dots, N-1$. Since $x_j x_j^\top$ has entries $\Delta^2/(4y_j^2)$, $-\Delta^2/4$, and $y_j^2\Delta^2/4$,

$$\Lambda = \frac{\Delta}{\sigma^2}\begin{pmatrix} A & -N \\ -N & B \end{pmatrix}, \qquad h = \begin{pmatrix} \dfrac{2}{\sigma^2}\,C + \dfrac{\Delta}{4}\,A \\[2ex] \dfrac{2}{\sigma^2}\,(B - D) - \dfrac{\Delta}{4}\,N \end{pmatrix}.$$

The terms $\frac{\Delta}{4}A$ and $-\frac{\Delta}{4}N$ both come from the Itô correction. The second looks negligible, being of order Δ, but it is summed over N steps; its total, $N\Delta/4$, grows with the length of the sample. By the Cauchy–Schwarz inequality $AB > N^2$ unless the path is constant, so Λ is positive definite. Inverting,

$$\Lambda^{-1} = \frac{\sigma^2/\Delta}{AB - N^2} \begin{pmatrix} B & N \\ N & A \end{pmatrix},$$

and the posterior means are

$$\mathbb{E}[\kappa\theta \mid \cdot\,] = \frac{2}{\Delta}\,\frac{BC + N(B-D)}{AB - N^2} + \frac{\sigma^2}{4}, \qquad \mathbb{E}[\kappa \mid \cdot\,] = \frac{2}{\Delta}\,\frac{NC + A(B-D)}{AB - N^2}. \tag{7}$$

The Itô correction moves the posterior mean of $\kappa\theta$ by exactly $\sigma^2/4$ and leaves the posterior for κ untouched.

Two refinements are easy. A proper Gaussian prior $\psi \sim N(m_0, V_0)$ replaces Λ by $\Lambda + V_0^{-1}$ and h by $h + V_0^{-1}m_0$. Positivity, $\kappa\theta > 0$ and $\kappa > 0$, or the Feller condition $2\kappa\theta \geq \sigma^2$, can be imposed by truncating the posterior, which in practice means discarding draws outside the region.

The shock scale σ needs no prior when the path is fine. Over a span of length t, the quadratic variation of y is $\sigma^2 t/4$, so

$$\hat{\sigma}^2 = \frac{4}{t}\sum_{j=0}^{N-1}(y_{j+1} - y_j)^2$$

is essentially exact on a fine grid.

4.4 The rest of a sampler

Equation 6 is one block of a data-augmentation sampler for discretely observed data (Eraker 2001). A complete sampler alternates three steps. It draws ψ given σ and the path, from Equation 6. It updates the imputed points given the observations and the parameters, by Metropolis–Hastings steps, for example one gap between observations at a time. And it updates σ, which is not conjugate because it enters both the drift and the error variance. The last step contains a trap. On a fine grid the imputed path pins down σ through its quadratic variation, so a sampler that alternates naively between σ and the imputed points mixes more and more slowly as M grows. Roberts and Stramer (2001) analyze this problem and remove it by reparameterizing the imputed path.

4.5 The stationary percentile, again

The posterior also gives an interval for the stationary percentile of Section 3 that respects the serial dependence of the data. The percentile is a function of $\alpha = 2\psi_1/\sigma^2$ and $\beta = 2\psi_2/\sigma^2$, so each posterior draw of ψ, combined with $\hat{\sigma}$, gives a draw of $q = F^{-1}(0.95;\, \alpha, \beta)$. The 2.5th and 97.5th percentiles of those draws form a 95% credible interval.

5 Simulation checks

The checks condition on the complete path, so they test the conjugate block of Section 4.3 in isolation. In a full sampler, the imputed points would be the sampler's current values.

First, an identity check. On one simulated path of 200,000 steps, the posterior mean computed from A, B, C, D, and N agrees with the least-squares fit of Equation 5 to within 1.3e-12, so the closed form and the regression are the same object.

Second, a sampling experiment. Each of 300 replications simulates 25 years of daily data with $M = 9$ imputed points per day ($\Delta = 1/2520$), using the exact transition of the CIR process (Cox et al. 1985), estimates σ from the quadratic variation, and computes the posterior Equation 6. The estimate $\hat{\sigma}$ averaged 0.2499, with a standard deviation of 0.0007 across replications. Table 1 reports the rest. If the posterior is right, the average posterior standard deviation should match the actual spread of the posterior means across replications, and the 95% intervals should cover the true values about 95% of the time.

Table 1: Posterior performance across 300 replications of 25 years of daily data. Intervals for $\kappa\theta$ and κ are posterior mean ± 1.96 posterior standard deviations; the interval for the stationary 95th percentile uses the 2.5th and 97.5th percentiles of posterior draws.

Quantity	True value	Mean of posterior means	SD of posterior means	Mean posterior SD	Coverage of 95% interval
$\kappa\theta$	0.080	0.085	0.013	0.013	96%
κ	2.00	2.15	0.41	0.41	95%
95th percentile of v	0.0879	0.0889	0.0118	0.0134	94%

The average posterior standard deviations match the actual spread of the posterior means, and coverage is close to the nominal 95%; with 300 replications, a coverage rate is measured to within about ± 2.5 percentage points (two Monte Carlo standard errors). The posterior mean of κ averages 2.15 against a true value of 2. This is the familiar upward bias of mean-reversion estimates in finite samples: 25 years of data contain only about $\kappa \times 25 = 50$ mean-reversion times, and the bias shrinks as the sample grows.

6 Scope

These notes treat the variance as observed at discrete dates. In Heston's model it is latent and only prices are observed, so a full treatment would add the price equation and update the variance path from prices as well. The conjugate block is checked in isolation, conditioning on the complete path; a full sampler would also update the imputed points and σ (Section 4.4). Finally, the Euler step introduces a discretization error that shrinks as M grows; the exact transition used in the simulations could replace it, at the cost of conjugacy.

Revision notes

These notes are a corrected write-up of handwritten derivations from 2024–25. Three results changed. The simulations above, rerun with the handwritten formulas, show what each error did.

The stationary law. The handwritten derivation gave Gamma$(1 + \kappa\theta/\sigma^2,\ \kappa/\sigma^2)$. It multiplied the drift into $\partial\varphi/\partial v$ instead of differentiating the product $\kappa(\theta - v)\varphi$, dropped the factor of one half on the diffusion term, and lost the product-rule term. For the running example it put the stationary mean at 0.071 rather than 0.040, and because its shape parameter exceeds one for every choice of parameters, it hid the Feller condition. It also approximated $\Gamma(\alpha)$ by Stirling's formula, which is unnecessary.

The percentile interval. The handwritten version used the normal approximation to the Gamma percentile. Its inverse Fisher information had the prefactor inverted and the diagonal signs reversed, which produced a negative variance for $\hat{a}$, and it combined the gradient with that matrix without the cross term. In the simulation of Section 3.1 its nominal 95% interval covered the true percentile in 49.9% of samples, against 94.8% for the corrected interval.

The conditional posterior. The handwritten expansion dropped the term $-\frac{\Delta}{4}N$ from h (in the notation of Section 4.3) and set every entry of the precision matrix to half its correct value. The first error pushes the posterior mean of κ toward 1.5κ in large samples; the second makes posterior standard deviations $\sqrt{2}$ times too large. In the sampling experiment of Section 5 the handwritten posterior mean of κ averaged 3.22 against a true value of 2, and its 95% intervals covered κ in 49% of replications and $\kappa\theta$ in 65%.

The write-up also renames the drift parameters $\psi = (\kappa\theta, \kappa)$, since the handwritten notes used α and β both for these and for the Gamma parameters, and replaces the term-by-term expansion with the regression form of Section 4.2, which gives the same posterior with less algebra.

References

Cox, John C., Jonathan E. Ingersoll, and Stephen A. Ross. 1985. "A Theory of the Term Structure of Interest Rates." *Econometrica* 53 (2): 385–407.

Eraker, Bjørn. 2001. "MCMC Analysis of Diffusion Models with Application to Finance." *Journal of Business & Economic Statistics* 19 (2): 177–91.

Feller, William. 1951. "Two Singular Diffusion Problems." *Annals of Mathematics* 54 (1): 173–82.

Heston, Steven L. 1993. "A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options." *Review of Financial Studies* 6 (2): 327–43.

Roberts, G. O., and O. Stramer. 2001. "On Inference for Partially Observed Nonlinear Diffusion Models Using the Metropolis–Hastings Algorithm." *Biometrika* 88 (3): 603–21.

Saporito, Angelo. 2026. *The Fokker–Planck Equation: Derivation and Applications in Economics and Finance*. Note, originally written May 2024, revised 2026.